# Software Engineering Lab

**Project Proposal Guidelines & Presentation Template**

Time Allocated: 10 Minutes (5–6 min presentation + 4–5 min Q&A)

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## 1. Overview & Core Objectives

The goal of the proposal presentation is to demonstrate that your project addresses a validated problem, presents a feasible technical solution, and has a clear plan for execution within the lab duration. Each team will have **10 minutes** total, structured strictly as **5–6 minutes for presentation** and **4–5 minutes for evaluation/Q&A**.

## 2. Detailed Proposal Sections

### Section 1: Problem Identification (Backed by Data)

- **Problem Definition:** State the problem clearly in simple terms. Avoid jargon here.
- **Importance & Relevance:** Why is this problem worth solving? Who is affected?
- **Evidence / Data:** Support your claims with basic statistics or verifiable facts (e.g., from research papers, news reports, industry surveys).
- **Real-World Impact:** Highlight the practical benefits (e.g., healthcare improvements, educational accessibility, efficiency gains, social benefit).

### Section 2: Solution (The Idea, Features, and Existing Systems)

- **The One-Line Elevator Pitch:** Describe your project in *exactly one sentence* that can be understood by non-technical audiences (e.g., a kid or your parents).
- **System Purpose:** Explain what your model/system will accomplish at a high level.
- **Core Features & Capabilities:** List the specific key actions the end system can perform.
  - *Example 1:* "Our system will transcribe medical conversations into structured clinical summaries."
  - *Example 2:* "Our chatbot will answer undergraduate course prerequisite questions instantly."
- **Existing Solutions & Differentiation:** Identify existing software, applications, or research solutions that address a similar problem. Explicitly highlight how your proposed system differs (e.g., better accuracy, localized data, lower cost, missing feature set, improved UI/UX).
- **Note:** Keep this section strictly idea-driven and user-focused; reserve technical mechanics for Section 3.

### Section 3: Plan of Action (High-Level Technology Overview)

- **Tech Stack & Rationale:** Specify key frameworks and tools (e.g., React, Python, TensorFlow, PyTorch, Scikit-learn, PostgreSQL) and briefly explain *why* each was chosen.

- **Libraries & Specialized Tools:** Mention relevant libraries for supporting tasks (e.g., Pandas for data manipulation, Matplotlib/Seaborn for visualization, NLTK/SpaCy for NLP).
- **Feature-wise Mapping:** Connect major features from Section 2 to their high-level technical implementation steps. Keep descriptions high-level (no low-level source code).

### Section 4: Project Timeline & Work Breakdown

- **Milestones:** Outline key phases (e.g., Data Collection & Preprocessing, Model Training / Core Backend, Frontend Development, Integration, Testing).
- **Existing Work / Current Progress:** Detail any work already completed (e.g., initial literature review, dataset sourcing, exploratory data analysis, environment setup, baseline experiments, or wireframes).
- **Schedule & Work Breakdown:** Use a Gantt chart or a clear week-by-week / sprint breakdown for remaining tasks.
- **Individual Task Allocation:** Explicitly specify **who will do what** across all major components (e.g., Member A: Data Pipeline & ML Model, Member B: Backend API, Member C: Frontend UI & Integration).

## 3. Recommended Presentation Structure (8–12 Slides)

| Slide # | Title | Content Focus | Est. Time |
|---|---|---|---|
| 1 | Title & Team | Project Title, Team Members, Student IDs, Roles | 0.5 min |
| 2 | Motivation & Context | Real-world context and why this domain matters | 0.5 min |
| 3 | Problem Identification | Core problem statement + verified statistics/data | 0.5 min |
| 4 | Proposed Solution | One-line elevator pitch + system overview | 0.5 min |
| 5 | Existing Systems | Current software analysis & key differentiators | 0.5 min |
| 6–7 | Core Features | Key end-user features & capabilities (1–2 slides) | 1.0 min |
| 8 | Technical Architecture | High-level system diagram & technology stack | 0.5 min |
| 9–10 | Feature Implementation | Mapping features to technological components | 1.0 min |
| 11 | Work Breakdown | Individual task allocations (**Who does what**) | 0.5 min |
| 12 | Timeline & Milestones | Gantt chart, sprint plan, project risks | 0.5 min |
| **Total** | | **Presentation Target** | **6.0 min** |

## 4. Presentation Checklist

- Is the elevator pitch understandable without prior technical domain knowledge?
- Is every problem statement backed by at least one factual data point or source reference?
- Have existing software tools been analyzed and clear differentiators/unique points highlighted?
- Are individual responsibilities (**who does what**) explicitly assigned for every feature/task?

- Is the slide deck structured between 8 and 12 slides and strictly within the 5–6 minute presentation limit?
- Have you rehearsed timing to ensure you leave 4–5 minutes for faculty Q&A?